

Financial Markets: 10. Real Estate

Robert J. Shiller

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Original lecture by Robert J. Shiller. Transcript-derived notes curated by [LazyingArt LLC](#).

Chapter 1

Real Estate

These notes follow Robert J. Shiller’s lecture as it unfolds in real time. We begin where he begins: with a roadmap. First comes the long history of mortgage lending; then commercial real estate finance; then residential real estate finance; and, if time permits, the larger architecture of securitization. That order matters. The lecture does not treat real estate finance as a list of institutions. It treats it as a chain of problems: first the existence of collateralized lending, then the legal preconditions for scaling it, then the ownership forms used in practice, then the redesign of the home mortgage, and finally the system-wide consequences of turning mortgages into securities.

1.1 Mortgage Lending in Long Historical Perspective

The lecture opens by widening the frame. Real estate finance, we are told, is about financial contracts that use real estate as collateral, and that subject is easier to understand if we refuse to start in the twentieth century. Shiller therefore begins with the word *mortgage*. He cautiously glosses an old Latin ancestor as a “death pledge,” and then notes that medieval French usage fed into the English word. The exact philology is not the main point here; the main point is that the institution is old.

The lecture then pushes the history much farther back than medieval England. Drawing on the work of Yale historian Valerie Hansen, Shiller points to trade documents from the Tang period and to Sogdian materials from the Silk Road world. Those records suggest that loans secured by collateral are more than a thousand years old. In some of the cases discussed in the lecture, the collateral could include property and even persons held in slavery, together with contractual obligations to maintain that collateral. The moral content is disturbing, but analytically it serves a purpose: the mortgage is not a modern invention.

That is why the lecture immediately introduces a contrast. Mortgage lending is ancient, but large-scale modern mortgage finance is not. The idea of collateral has been around for a very long time. What arrived much later was the institutional environment that could support a broad and trusted mortgage market. That distinction carries us into the next step.

1.2 Property Rights as the Precondition for Mortgage Markets

If mortgages are so old, why did modern mortgage markets develop so late and so unevenly? Shiller's answer is that the bottleneck was not conceptual novelty. It was legal and administrative clarity.

The lecture's most vivid example is a Hartford Courant advertisement from 1778. Elisha Cornwell sold his farm on mortgage terms for 800 pounds. The buyer did not fully pay him, yet proceeded to mortgage the same farm again, first for 880 pounds and later for 1,000 pounds. Cornwell had to warn later victims in the newspaper. That detail is important. It shows just how underdeveloped mortgage institutions were. The problem was not merely that one man behaved badly. The deeper problem was that there was no systematic public way to determine who actually owned the farm.

So we should not summarize the point too quickly as "property rights matter." The lecture is more concrete than that. Mortgage lending requires a title system that lets strangers verify ownership, and it requires foreclosure rules that let lenders recover the collateral if the borrower does not pay. Without those two things, the contract cannot be scaled.

That is why Shiller dwells on the Prussian *Grundbuch*. In 1872, Prussia created a central book of ownership, and in 1897 the institution became national within Germany. The lecture uses that example to mark a turning point: once the state provides a reliable registry, mortgage lending can move beyond local trust networks and become a national financial business.

The argument then broadens again. Hernando de Soto's *The Mystery of Capital* is brought in not as a side citation but as evidence that the same problem persists in much of the developing world. One may be able to go to a village and ask around who owns a parcel of land, but a serious financial transaction cannot rest on village hearsay. Nor can mortgage lending flourish if foreclosure takes years or if courts do not function well enough to make collateral recovery credible.

1.2.1 Question & Answer

Question. Why can mortgage lending not rest on local knowledge or hearsay about ownership?

Answer. Because a mortgage is not merely a social understanding. It is a priced contract. The lender must know that the borrower really owns the property and that this fact can survive dispute, resale, and enforcement. If ownership is known only informally, the collateral is unreliable. If foreclosure is legally possible on paper but practically impossible in court, the collateral is again unreliable. In that case the lender will either refuse to lend or charge terms severe enough to make the market thin. The lecture's claim is therefore stronger than a slogan: mortgage markets scale only when ownership and repossession become legible and enforceable.

There is a hard institutional symmetry here. It may seem cruel to throw someone out of a house after default, and the lecture acknowledges that moral discomfort. But Shiller immediately adds the lender's side: if the house cannot be recovered, no one will make the mortgage in the first place. That is part of the same development story. The growth of mortgage finance in the twentieth and twenty-first centuries depends on improvements in both title and enforcement.

1.3 Commercial Real Estate Before Democratization

With that long history in place, the lecture narrows to the United States. Shiller tells us explicitly that finance tends to standardize within countries: laws, regulators, and public expectations push people toward a small number of accepted contract forms. Innovation is therefore slow, and this is already one of the lecture's recurring themes.

He then asks a practical question: who owns the commercial buildings around us? The answer is often not a public corporation. In the lecture, the typical answer is a partnership.

The reason is tax treatment. Corporations are subject to corporate tax, and their owners are taxed again on the income they receive. Partnerships allow income to pass through more directly. If one owns a building through a partnership rather than through a corporation, one can avoid this double taxation. That is why so much stand-alone commercial real estate has been organized in partnership form.

The lecture gives that partnership form a more specific name: the direct participation program, or DPP. A DPP is direct in the sense that the investor participates directly in the profits of the real-estate venture as a partner rather than as a shareholder. It is typically not offered to the general public. Instead, it is restricted to accredited investors, whom the lecture describes roughly as wealthy investors meeting SEC wealth or income thresholds.

Three structural features matter in the lecture's account. First, the DPP is usually project-specific. Second, it has a limited life rather than perpetual duration. Third, it often distinguishes between general partners and limited partners. The general partner manages the business and lacks limited liability in the strongest sense; the limited partners are passive investors and enjoy limited liability.

That structure fits a building rather well. A group can acquire or develop a property, manage it for a period, depreciate it over the life of the venture, and then sell it and wind the vehicle down. The lecture's local example is 360 State Street in New Haven: Shiller does not claim to know its actual financing structure, but uses it as a plausible illustration of how a real-estate partnership could be organized around one building, with a plan for operation and an eventual exit.

So this part of the lecture does more than describe a niche legal form. It reveals a tension that will matter later. The tax-efficient structure for commercial real estate is also a structure that excludes most ordinary investors.

1.4 REITs and the Democratization of Commercial Real Estate

That tension is the entry point for REITs. Shiller explicitly frames them as part of the democratization of finance.

The political complaint was easy to state. If partnerships and DPPs are tax-efficient, why should access be limited to accredited investors? Why should most households be confined to ordinary corporate forms while a narrower wealthy group receives the better tax treatment available in commercial real estate? Congress's answer, in 1960, was to create the real estate investment trust, or REIT.

The lecture presents the REIT as a compromise. It opens real-estate investing to the broader public, but only by imposing a narrow legal definition of what the vehicle may do. In the lecture's compressed summary, a REIT must satisfy a set of threshold conditions:

- at least 75% of assets must be in real estate or cash;
- at least 75% of income must come from real estate;
- at least 90% of income must come from real-estate dividends, interest, and capital gains;
- at least 95% of income must be paid out;
- no more than 35% of income may come from sales of properties held for less than four years.

The point of these thresholds is not just definitional neatness. Once Congress creates a vehicle that avoids ordinary corporate taxation, every company would like to qualify. So the state confines the privilege to a carefully delimited real-estate form. One should read these figures here as lecture-era statutory conditions, not as a polished current-law summary.

1.4.1 Question & Answer

Question. If partnerships avoid double taxation, why not simply let everyone invest through partnerships directly?

Answer. Because the government does not want the partnership form to become a universal substitute for the corporation. The partnership form is therefore hemmed in by restrictions: limited life, project-specific structure, access limits, and distinctions between active and passive partners. The REIT is a different compromise. It extends access to small investors, but only by binding the vehicle to real estate, payout rules, and other statutory constraints. In the lecture's terms, it is democratization, but democratization through a regulated legal form.

Shiller then slows us down again with a historical theme he clearly cares about: innovation in finance is slow. REITs are created in 1960, but they do not immediately dominate the market. The lecture describes a succession of booms. The first comes in the late 1960s, when regulated deposit-rate ceilings make REIT payouts comparatively attractive. Another comes after the tax reform of 1986, which removes some of the tax attractions of DPPs and partnerships. A third expansion comes with the real-estate boom of the 1990s, when specialized REITs become more visible and the form finally becomes broadly familiar.

That long gestation matters because the lecture is making two parallel claims. One is that innovation is slow and path-dependent. The other is that finance tends, over long stretches of time, toward wider participation.

1.5 The Great Depression and the Redesign of the Home Mortgage

Only after commercial real estate has been organized in these terms does the lecture pivot to residential real estate. The pivot is motivated by scale. Housing is socially larger and financially more pervasive than commercial real estate. In the United States, the lecture gives a rough homeownership rate of about two-thirds, and treats that fact as partly the product of public policy that encourages mortgage lending.

To explain why the American home mortgage took its modern form, the lecture turns to the Great Depression. This is not just historical background. It is the crisis that exposes a flaw in the earlier contract design.

Before the Depression, mortgages in the United States were often short, typically two to five years, and they were balloon-payment mortgages. The lecture's arithmetic example is:

$$H_0 = \$10,000, \quad B_0 = \$5,000, \quad D_0 = \$5,000, \quad (1.1)$$

where H_0 is the house value, B_0 the mortgage balance, and D_0 the down payment. The borrower pays monthly interest, but the principal is largely untouched until maturity:

$$x_t = \frac{r}{12} B_0 \quad \text{for } t = 1, \dots, T-1, \quad B_T = B_0. \quad (1.2)$$

The whole arrangement depends on refinancing. When the balloon date arrives, the borrower expects to return to the bank and roll the loan.

The Depression breaks that expectation. Unemployment rises to about 25%. House prices fall sharply, often by more than half. In the lecture's example, a house bought for \$10,000 with a \$5,000 mortgage may fall to about \$4,000, so that

$$H_1 \approx \$4,000 < B_0 = \$5,000. \quad (1.3)$$

Now the borrower arrives at the refinancing date with exactly the wrong balance sheet: weak income, weak collateral, and a full principal balance still due. The bank refuses renewal. The family is then pushed toward bankruptcy or distress sale, and the original down payment is lost. The lecture is careful here. The problem is not merely that borrowers stopped paying. The deeper problem is that the contract hid its fragility in the refinancing date.

This is why the Roosevelt administration treats the old mortgage form itself as defective. The Federal Housing Administration, created in 1934, begins insuring mortgages. Mortgage insurance means that if the borrower defaults and the house value is insufficient, the government makes up the lender's shortfall. At the same time, FHA-insured mortgages are required to be long-term and non-balloon. The lecture first emphasizes 15-year amortizing mortgages, and then notes that by the early 1950s the FHA is pressing toward 30-year mortgages.

Shiller pauses over the word *amortize*: it means to pay down the balance. An amortizing mortgage therefore has no giant payment at the end. The family makes a fixed monthly payment, and when the term is over, the balance is zero. That is the conceptual payoff of the crisis story. The redesign is not cosmetic. It moves the risk out of a catastrophic lump-sum date and spreads it over a long, predictable payment stream.

1.6 The Mathematics of the Fixed-Rate Amortizing Mortgage

At this point the lecture deliberately slows down. Shiller says that the arithmetic of the amortizing mortgage can be confusing, and that in the 1930s it took real explanation. This is the mathematical center of the chapter.

We take the notation directly from the lecture. Let m be the maturity in months, let r be the annual mortgage rate, let B_0 be the initial mortgage balance, and let x be the fixed monthly payment. The contractual condition is that the present value of all monthly payments equals the amount initially borrowed. In clean notation, the lecture's formula is

$$B_0 = \sum_{t=1}^m \frac{x}{(1 + r/12)^t} \quad (1.4)$$

$$= x \sum_{t=1}^m (1 + r/12)^{-t} \quad (1.5)$$

$$= x \frac{1 - (1 + r/12)^{-m}}{r/12}. \quad (1.6)$$

This is simply the annuity formula, written with the monthly discount rate $r/12$. Shiller briefly self-corrects in the lecture after an oral slip; the correct rate here is the monthly rate $r/12$, and that is the form we keep.

Solving for the payment gives

$$x = B_0 \frac{r/12}{1 - (1 + r/12)^{-m}}. \quad (1.7)$$

That tells us the constant monthly payment. But the lecture is equally interested in something subtler: the internal composition of that payment. Each month's payment splits into interest and principal:

$$x = I_t + P_t, \quad (1.8)$$

$$I_t = \frac{r}{12} B_{t-1}, \quad (1.9)$$

$$P_t = x - I_t, \quad (1.10)$$

$$B_t = B_{t-1} - P_t. \quad (1.11)$$

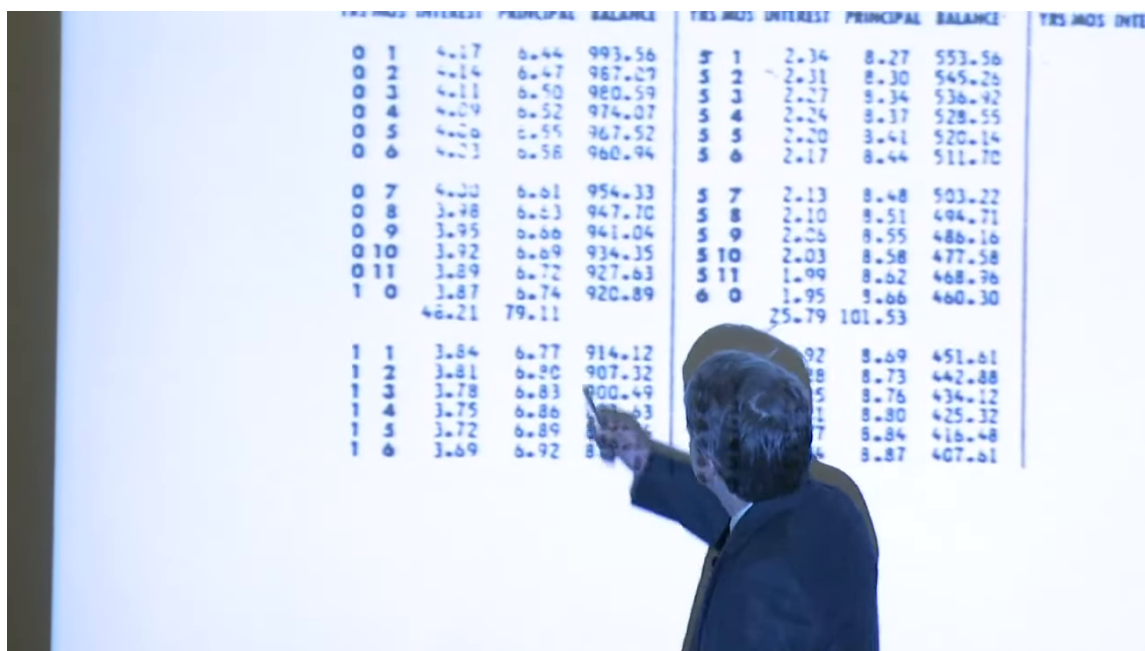
This recursion is the essential mechanism. Interest is computed on the remaining balance B_{t-1} . As the balance falls, interest falls. Since the total payment x is fixed, the principal component must rise. The household sees a stable monthly bill, but internally the contract is steadily shifting from interest to amortization.

The lecture then produces visual evidence from an old mortgage table. Before computers, lenders relied on printed tables to carry out exactly this arithmetic.

The projected page is partly blurred, but its structure is clear. The headers YRS, MOS, INTEREST, PRINCIPAL, and BALANCE repeat across the page so that a long schedule can be printed compactly. The left side shows early months, when interest is relatively high and the balance is still close to the original loan amount. The right side shows later months, where interest is smaller, principal is larger, and the balance has fallen substantially. Shiller's pointing gesture is part of the evidence here: he is drawing attention to a later row precisely to show the declining interest share.

The lecture's worked example uses a 10-year mortgage at 5%, with the table scaled per \$1,000 of initial balance. Thus

$$r = 0.05, \quad m = 120, \quad B_0 = \$1,000, \quad x \approx \$10.61. \quad (1.12)$$



TIME	INTEREST	PRINCIPAL	BALANCE	TIME	INTEREST	PRINCIPAL	BALANCE	TIME	INTEREST	PRINCIPAL	BALANCE
0 1	4.17	6.44	993.56	5 1	2.34	8.27	553.56	10 1	1.99	8.62	468.76
0 2	4.14	6.47	987.09	5 2	2.31	8.30	545.26	10 2	1.96	8.65	460.30
0 3	4.11	6.50	980.59	5 3	2.27	8.34	536.92	10 3	1.93	8.68	451.61
0 4	4.09	6.52	974.07	5 4	2.24	8.37	528.55	10 4	1.90	8.71	442.88
0 5	4.06	6.55	967.52	5 5	2.20	8.41	520.14	10 5	1.87	8.74	434.12
0 6	4.03	6.58	960.94	5 6	2.17	8.44	511.70	10 6	1.84	8.77	425.32
0 7	4.00	6.61	954.33	5 7	2.13	8.48	503.22	10 7	1.81	8.80	416.48
0 8	3.98	6.63	947.70	5 8	2.10	8.51	494.71	10 8	1.78	8.83	407.61
0 9	3.95	6.66	941.04	5 9	2.06	8.55	486.16				
0 10	3.92	6.69	934.35	5 10	2.03	8.58	477.58				
0 11	3.89	6.72	927.63	5 11	1.99	8.62	468.76				
1 0	3.87	6.74	920.89	6 0	1.95	8.66	460.30				
	46.21	79.11			25.79	101.53					
1 1	3.84	6.77	914.12								
1 2	3.81	6.80	907.32								
1 3	3.78	6.83	900.49								
1 4	3.75	6.86	893.63								
1 5	3.72	6.89	886.74								
1 6	3.69	6.92	879.82								

Figure 1.1: Historical amortization table showing how a fixed mortgage payment shifts from interest toward principal over time.

The first month is computed directly from the update rule:

$$I_1 = \frac{0.05}{12} \cdot 1000 \approx \$4.17, \quad (1.13)$$

$$P_1 = 10.61 - 4.17 = \$6.44, \quad (1.14)$$

$$B_1 = 1000 - 6.44 = \$993.56. \quad (1.15)$$

The second month repeats the same logic on the new balance:

$$I_2 = \frac{0.05}{12} \cdot 993.56 \approx \$4.14, \quad (1.16)$$

$$P_2 = 10.61 - 4.14 = \$6.47, \quad (1.17)$$

$$B_2 = 993.56 - 6.47 = \$987.09. \quad (1.18)$$

That is already enough to see the pattern. The balance has fallen, so interest has fallen from \$4.17 to \$4.14. The total payment stays at \$10.61, so principal rises from \$6.44 to \$6.47.

A compact reconstruction of the table makes the progression explicit:

The later checkpoint is where the lecture draws a practical lesson. After about six years and six months, the borrower has not paid off the whole mortgage, but the remaining balance is only about \$407.61. That is why the table matters. People often sell the house before the mortgage matures, and then one needs to know the unpaid balance immediately. The table answers that question.

YRS	MOS	INTEREST	PRINCIPAL	BALANCE
0	1	4.17	6.44	993.56
0	2	4.14	6.47	987.09
0	3	4.11	6.50	980.59
0	4	4.09	6.52	974.07
6	6	≈ 1.74	≈ 8.87	≈ 407.61

Table 1.1: A compact reconstruction of the lecture’s 10-year, 5% amortization example per \$1,000 borrowed. The early rows are visible in the projected table; the later checkpoint follows the lecture’s spoken arithmetic and should be read cautiously because the projected page is partly obscured.

1.6.1 Question & Answer

Question. If the monthly payment is fixed, why does the interest share fall while the principal share rises?

Answer. Because the contract computes interest on the remaining balance, not on the original balance. Early in the mortgage, the remaining balance is large, so the interest charge is large. Later in the mortgage, the balance is smaller, so the interest charge is smaller. Since the total payment x is fixed, every fall in I_t automatically shows up as a rise in P_t . This is the core time structure of the amortizing mortgage: a constant total payment with a changing internal composition.

That changing composition is what makes the fixed-rate amortizing mortgage both safer and easier to live with. The family no longer faces a giant balloon date. Instead, it faces a predictable monthly obligation, while the contract quietly pays itself down in the background.

The lecture now gives the instrument its standard name: the conventional fixed-rate mortgage. But Shiller does not stop there. He immediately turns success into a puzzle. Why did such a sensible contract become dominant in so few countries?

1.7 From Fixed-Rate Mortgages to Securitization, Crisis, and Reform

The lecture’s first answer to the international puzzle is that the fixed-rate mortgage is costly to supply. If a lender guarantees a rate for 20 or 30 years, that promise has value and must be priced. Borrowers may resist the higher rate. Regulators may also resist the maturity mismatch it creates: banks fund themselves with deposits that can leave quickly, but the mortgage asset stays on the books for decades. In a run, the bank cannot liquidate that asset cheaply. So the long fixed-rate mortgage is not just a household contract. It is a balance-sheet problem and a regulatory problem.

Shiller uses other countries to sharpen that point. Canada, for example, did not make the long-term fixed-rate mortgage standard in the same way as the United States. He mentions a Canadian episode around 1980, when rates rose sharply and refinancing became painful for many households. He also mentions Alastair Darling’s call for longer-term mortgages in the United Kingdom after the crisis. The underlying theme is that the American mortgage form, however useful, is institutionally contingent.

The lecture then turns from the mortgage contract itself to the state-supported market that grew

around it. In 1938, the federal government created what became known as Fannie Mae, initially as a government agency that bought mortgages in order to support lending and stimulate the depressed housing sector. The economic logic is simple: when a bank makes a mortgage and then sells it, it gets cash back and can lend again. A secondary market therefore supports the primary market.

Later Freddie Mac is created. Shiller presents Freddie as initially associated more explicitly with mortgage securitization: buying mortgages, packaging them into securities, and selling them with a guarantee. Over time both Fannie and Freddie operate in that space. So the mortgage moves one step farther away from the household. It becomes the raw material of tradable securities sold to investors.

1.7.1 Question & Answer

Question. Why were Fannie Mae and Freddie Mac rescued after repeated claims that they were not government-guaranteed?

Answer. Because the lecture's distinction between explicit and implicit guarantees collapses in crisis. Formally, officials could say that the government did not stand behind these private corporations. But investors around the world treated them as government-linked, and the United States benefited from that belief. Once the crisis of 2008 arrived, letting them fail would have damaged not only mortgage markets but also the perceived safety of U.S. financial commitments more broadly. The legal disclaimer did not survive the reputational and systemic cost of enforcing it.

This is one of the lecture's most memorable institutional lessons. One can deny a guarantee repeatedly. But if the state has created the institution, allowed the market to rely on it, and let the exposure become enormous, the denial may not be credible when the crisis finally arrives.

The lecture briefly compares Canada again at this point. Canada has a government-owned housing-finance institution, the Canada Housing and Mortgage Corporation, which does some work analogous to the FHA and the U.S. secondary-market agencies. But Shiller's point is that Canada did not let that support structure become as large or as aggressively expansionary as the United States did. That difference helps explain why the Canadian housing bubble was less extreme.

The final movement of the lecture broadens into the full securitization chain. The textbook, Shiller says, discusses securities such as collateralized mortgage obligations (CMOs) and collateralized debt obligations (CDOs). A CMO is a mortgage-backed structure that divides cash flows into tranches, especially by prepayment exposure. A CDO is a further layer, often holding mortgage-linked debt instruments as assets. By the time we reach this stage, the original home mortgage sits inside a much more complicated financial machine.

The lecture insists on the sequence of intermediaries. Someone originates the mortgage by dealing directly with the homeowner. That mortgage is then sold to an investor or securitizer. A servicer may collect the payments and deal with delinquency. The mortgage pool can then be reorganized into CMOs, and those securities can in turn become inputs into CDOs. What appears at the household level as one mortgage has become, at the system level, a chain of layered claims held all over the world.

This is where the lecture places the moral-hazard problem. If the originator knows the mortgage will be sold immediately, the originator may become cynical about repayment quality. Shiller gives

a blunt example: a mortgage broker may encourage a family to overstate income because the broker expects to pass the mortgage on and avoid the loss. That is why the post-crisis reform he highlights is so simple in form and so revealing in purpose. If M denotes a mortgage pool, then the rule is

$$\text{originator retains } 0.05M, \quad \text{originator sells } 0.95M. \quad (1.19)$$

The idea is to keep some risk with the party that screened the borrower. Europe moved in that direction, and the United States incorporated the same 5% principle into Dodd–Frank. The lecture also notes related professionalization, such as stricter licensing of mortgage brokers.

The ending is deliberately guarded. Shiller does not say that we should abolish securitization and go back to 1778. He says instead that the basic system has been retained, while reforms try to reduce the worst incentive failures. That is a more modest conclusion, but it is also a more faithful one.

1.8 Summary

The lecture moves in a deliberate arc. Mortgage lending is ancient, but broad mortgage markets are modern because they depend on clear title, enforceable collateral, and administratively legible ownership. Commercial real estate then shows us how tax law and regulation shape the ownership form, first through partnerships and DPPs and later through REITs as a democratizing compromise. Residential finance raises the stakes: the Great Depression reveals that the old balloon mortgage hid a dangerous refinancing risk, and the FHA-backed shift to long-term amortizing mortgages redesigns the household contract. The mathematical core of that redesign is an annuity-pricing equation together with a simple monthly recursion for interest, principal, and balance. Finally, the lecture zooms back out. The fixed-rate mortgage becomes an international puzzle, then the raw material for securitization, implicit guarantees, global investment, crisis, and reform. The result is a chapter in which a household payment stream and a world financial system turn out to be two views of the same contract.